

XIAOYING LIAO

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EDUCATION

Johns Hopkins University (JHU)

Aug. 2024 – May. 2025

Master of Science in Engineering (M.S.E.) in Biomedical Engineering

Baltimore, USA

- Supervised by *Professor Carlo Iomini*, **GPA:** 4.00/4.00
- Courses:** Cellular Engineering, Tissue Engineering, Cell & Tissue Engineering Lab, Advanced Topics in Regenerative and Immune Engineering, Human Genetics, Systems Pharmacology and Personalized Medicine

University of Sydney (USYD)

Feb. 2020 – Dec. 2023

Bachelor of Science (Honours) in Statistics

Sydney, Australia

- Supervised by *Professor Jean Yang*, **GPA:** 3.90/4.00, **Rank:** 1/200
- Courses:** Probability and Estimation Theory, Statistical Inference, Stochastic Process, Advanced Statistical Modelling, Statistical Machine Learning, Life and Evolution

SKILLS & HONORS

Software Tools: Python, R, SQL, MATLAB, GraphPad Prism, Stata, SAS, SPSS, Tableau, FIJI/ImageJ, BLAST/Database Search, Microsoft Office Set; Proficient in Single-Cell and Spatial Transcriptomics Analysis

Wet Lab Skills: Cell and tissue culture; Immunostaining; Confocal and two-photon imaging; Gene delivery and transfection; PCR and qRT-PCR; Western blot; Genotyping; DNA/RNA extraction; Gel electrophoresis; Cell apoptosis; Tissue modeling; Metabolic glycoengineering; ORO staining; Mouse handling and procedures (tamoxifen injection, tissue dissection)

Honors: Vice Chancellor's Global Mobility Scholarship (2020), Dalyell Scholar (2022), 2nd Prize, China National High School Biology Olympiad (2018)

PUBLICATIONS

Python Song, **Xiaoying Liao**. *ProactiveEngine: Proactive Learning with Lifelong Memory via World Models*. Empirical Methods in Natural Language Processing (EMNLP, Main Conference), 2026.

Oral (Top 1%)

Miao Cui, Cinlong Huang, **Xiaoying Liao**, Xiaoke Song, Stephanie Conley, Jesse Engreitz, Xiaojie Qiu. *Spatio-temporal Transcriptomic Profiling of Serial Human Fetal Heart Sections Reveals Cellular Architecture and Functional Niches during development*. Stanford Genetics Department Retreat, 2026. Poster presentation.

Huanhuan Xiao, **Xiaoying Liao**, Céline Portal, Zhenzhen Pan, Eileen Chen, Igor A. Butovich, Carlo Iomini. *Hedgehog Signaling Maintains Meibomian Gland Progenitors and Lipid Homeostasis with Implications for Meibomian Gland Dysfunction Therapeutics*. Investigative Ophthalmology & Visual Science, 2026. pp. 3883.

Nianping Liu, Xuanzhi Chen, Miao Cui, **Xiaoying Liao**, Xiaoke Song, Sairam Pantham, Weize Xu, Xiaojie Qiu. *Bio-Babel: Autonomous Cross-Language Reconstruction of Agent-Ready Computational Biology Ecosystems*. bioRxiv, 2026.

Tianyang Wang, Xi Xiao, **Xiaoying Liao**. *Foundation Models for Radiology Report Generation*. One chapter in medical field's book, 2026. Under review.

Xi Xiao, Yunbei Zhang, Lin Zhao, Yiyang Liu, **Xiaoying Liao**, Zheda Mai, Xingjian Li, Xiao Wang, Hao Xu, Jihun Hamm, Xue Lin, Min Xu, Qifan Wang, Tianyang Wang, Cheng Han. *Prompt-based Adaptation in Large-scale Vision Models: A Survey*. Transactions on Machine Learning Research, 2026.

Junjie Li, Xi Xiao, Yunbei Zhang, Chen Liu, Lin Zhao, **Xiaoying Liao**, Yingrui Ji, Janet Wang, Jianyang Gu, Yingqiang Ge, Weijie Xu, Xi Fang, Xiang Xu, Tianchen Zhao, Youngeun Kim, Tianyang Wang, Jihun Hamm, Smita Krishnaswamy, Jun Huan, Chandan K. Reddy. *Agent Harness Engineering: A Survey*. Under review at ACL, 2026.

Tianyang Wang, Xi Xiao, Gaofei Chen, **Xiaoying Liao**, Guo Cheng, Yingrui Ji. *Boosting Active Learning with Knowledge Transfer*. 2025 IEEE 37th International Conference on Tools with Artificial Intelligence, 2025. pp. 1124–1131.

Xiaoying Liao. *Assessing the Performance of Clustering Methods in Spatially Resolved Transcriptomics Data*. University of Sydney, 2023. Undergraduate honors research.

Manuscripts in Preparation

Huanhuan Xiao, Xiangqian Shi, Abi Cha, Emma Liu, Randi Marshall, **Xiaoying Liao**, Molly Voit, Uzoamaka Nwagbo, Frazer Matthews, James H. Thierer, Meera Saxena, Ximin Li, Tae-In Kam, Xiangrong Kong, Jeff S. Mumm, Liyun Zhang. *Novel zebrafish genetic models of pde6b mutants exhibit rod photoreceptor degeneration*. Manuscript completed; planned submission to Communications Biology by October 2026.

Huanhuan Xiao, **Xiaoying Liao**, Céline Portal, Zhenzhen Pan, Eileen Chen, Igor A. Butovich, Carlo Iomini. *Genetic interaction between primary cilia and Hedgehog signaling regulates meibomian gland development*. Manuscript in preparation; expected submission by December 2026.

Huanhuan Xiao, Liyun Zhang, **Xiaoying Liao**, Emma Liu, Shaojun Du. *Expression and functional analysis of Hspa8 in early zebrafish development*. Manuscript completed; planned submission to Cells in 2026.

RESEARCH EXPERIENCE

Ciliary Hedgehog Signaling in Meibomian Gland Development and Dysfunction Oct. 2024 – Present

Research Assistant supervised by Professor Carlo Iomini

JHU, Baltimore

- **Age-associated decline in Hedgehog signaling:** Analyzed transcriptomic profiles of tarsal plates from young and aged human donors, identifying reduced expression of Hedgehog target genes (*Gli2*, *Ptch1*) and stem cell-associated markers in aged meibomian glands.
- **Genetic dissection of ciliary Hedgehog signaling:** Used K14-Cre-based conditional knockout mouse models (*Smo*, *Ift88*) to define stage-specific roles of primary cilia and Hedgehog signaling in meibomian gland development, showing that cilia-dependent Hh signaling is essential for gland morphogenesis and adult tissue homeostasis.
- **Partial rescue of gland atrophy in double mutants:** Found that additional deletion of *Ift88* in *Smo* mutants partially alleviates meibomian gland degeneration, with transcriptomic changes suggesting compensatory involvement of signaling pathways.

Validating *Gne* gene Knockout CHO-K1 Cell Line

Sep. 2024 – Dec. 2024

Cell & Tissue Engineering Lab Advanced Project Leader

Baltimore, USA

- **Validation of CRISPR-mediated gene knockout:** Confirmed the successful deletion of *Gne* exons 2 and 4 in CHO-K1 cells using PCR and Sanger sequencing, verifying intended one-base deletion/insertion and precise editing outcomes.
- **Functional knockout confirmation at the protein level:** Conducted Western blot to assess Gne protein expression; observed absence of Gne signal in knockout cells while GAPDH served as loading control, supporting transcriptional-level findings.

Assessing the Performance of Clustering Methods in Spatial Transcriptomics Feb. 2023 – Dec. 2023

Supported by an Australia National Health and Medical Research Council (NHMRC) Grant (1173469)

USYD, Sydney

- **Data Preprocessing:** Utilized Python and R to conduct thorough data preprocessing on spatial transcriptomics datasets, achieving data integrity through quality control, normalization, feature selection, and PCA-driven dimensionality reduction.
- **Spatial Domain Detection Analysis:** Applied diverse clustering techniques, including K-means and Seurat, and innovative spatial clustering methods like BayesSpace and STitch3D, enhancing the understanding of tissue structure in biological studies.
- **Impact Factor Study:** Evaluated clustering method efficacy across simulation and real datasets, revealing critical insights that guide further spatial transcriptomics research.

Quantitative Evaluation of Benchmarking in Single-Cell Analysis

Feb. 2023 – June 2023

Supported by the Sydney Precision Data Science Centre and the World Health Organization (WHO)

USYD, Sydney

- **Developed a structured framework for literature evaluation:** Designed a comprehensive questionnaire with over 70 questions to assess all aspects of single-cell benchmarking research. Conducted a broad review to evaluate single-cell research, screening 245 pertinent studies from 563, facilitating a broad perspective on the field.
- **Statistical Modeling and Analysis:** Implemented varied regression analyses to discern critical factors in single-cell benchmarking, endorsing the use of multiple linear regression for robust predictive insights.
- **Differentiate Research Categorization:** Applied logistic regression in categorizing research papers into benchmarking versus method development. Revealed through data analysis that more method comparisons, speed measurements, and lower impact factors characterize pure benchmarking research papers.

Type 2 Diabetes Risk Prediction and Food Intake Recommendations

Aug. 2022 – Dec. 2022

Capstone Project Leader, STAT3888 Statistical Machine Learning

USYD, Sydney

- **Feature Selection:** Calculated correlations between features and HbA1c (Hemoglobin A1c), then identified key variables and created new features better to represent lifestyle and dietary influences on HbA1c levels.
- **Regression Analysis:** Developed robust linear, ridge, and Lasso regression models to evaluate the influence on HbA1c indicator, enhancing disease risk prediction accuracy.
- **Decision Tree Modeling :** Selected the best decision tree model through detailed cross-validation and accuracy comparison, then identified high-risk individuals using selected variables and provided personalized nutritional advice and health interventions to reduce HbA1c levels and improve health.

TEACHING EXPERIENCE

Johns Hopkins University <i>Teaching Assistant</i> • Cell & Tissue Engineering Lab: Mentored students in practical modules on cell culture, gene delivery, metabolic glycoengineering, and tissue modeling. Provided support in data collection, data analysis, and scientific lab reports.	Jan. 2025 – Present <i>Baltimore, United States</i>
University of Sydney <i>Teaching Assistant</i> • STAT2011 Probability and Estimation Theory & DATA2002 Data Analytics: Led tutorials and workshops on probability, statistical inference, R programming, and introductory machine learning for real-world data analysis.	Feb. 2023 – Dec. 2023 <i>Sydney, Australia</i>

WORKING EXPERIENCE

Johns Hopkins University School of Medicine <i>Research Assistant, supervised by Professor Carlo Iomini, Wilmer Eye Institute</i> • Project Contribution: Contributed substantially to a project studying cilia-dependent Hedgehog signaling in meibomian gland development, progenitor maintenance, lipid homeostasis, and dry eye disease. • Mechanistic Findings: Characterized a duct-associated Hedgehog-responsive epithelial progenitor program required for meibomian gland morphogenesis, epithelial proliferation, meibocyte maturation, and lipid production. • Functional and Collaborative Analysis: Demonstrated that <i>Ift88</i> ablation partially rescues Smo-dependent meibomian gland defects, including progenitor marker expression, epithelial proliferation, gland morphology, lipid production, and lipid composition, and supported collaborative work involving lipidomic profiling, transcriptomic analysis, and quantitative imaging, and dry eye disease-focused translational research.	Sep.2024 – Present <i>Baltimore, MD</i>
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ACADEMIC ACTIVITIES

Second author on a poster presented at Association for Research in Vision and Ophthalmology (ARVO)	2026
Second author on a poster presented at Johns Hopkins Cell Biology Symposium	2026
Second author on a poster presented at the Annual Wilmer Research Meeting	2025
Participant, Sydney Bioinformatics Research Symposium (SBRS)	2023
Participant, Unlocking Single Cell Spatial Omics Analyses with Scdneq	2023